

Global Urbanization & Population Density

1. Objective / Problem Statement

For this capstone, I set out to apply the full data analytics workflow, from sourcing and cleaning data through to analysis, visualization, and reporting, to a real-world development problem: understanding how urbanization and population density have changed globally between 2000 and 2024.

Using the World Bank's World Development Indicators (WDI) dataset, I looked at how fast the world is urbanizing, the pressure that growing population density places on cities, and how these patterns differ across regions and income groups. My goal was to turn raw indicator data into insights that could genuinely support planning and policy decisions, for stakeholders such as national governments, urban planners, and international development bodies.

2. Dataset Description

The World Development Indicators (WDI) is the World Bank's flagship collection of global development statistics, covering over 200 countries and territories across themes including economy, health, education, environment, and governance.

I worked with three files from the WDI bulk download:

- **WDICSV.csv**, the core dataset, containing indicator values for every country and year from 1960 to 2025.
- **WDISeries.csv**, the indicator catalogue, used to identify and select relevant indicators by topic.
- **WDICountry.csv**, country metadata, providing Region and Income Group classifications used for comparison.

The theme Environment: Density & Urbanization. Was chosen for this project.

From there, I selected six indicators to build the analysis around:

Indicator	Code	Missing Data (2000-2024)
Population density (people per sq. km)	EN.POP.DNST	~2%
Urban population (% of total)	SP.URB.TOTL.IN.ZS	~0.4%
Urban population growth (annual %)	SP.URB.GROW	~2%
Rural population (% of total)	SP.RUR.TOTL.ZS	~0.4%
Population in cities >1M (% of total)	EN.URB.MCTY.TL.ZS	~40%
Slum population (% of urban)	EN.POP.SLUM.UR.ZS	~63%

Once cleaned, my final analysis dataset covered 217 countries across a 25-year period, from 2000 to 2024.

3. Data Cleaning Process

I used two tools for the cleaning of the csv files.

3.1 Excel: indicator selection

I started in Excel reviewing WDI Series.csv (the indicator catalogue), which was to browse indicators by topic and narrow down the six most relevant to my Density and Urbanization theme.

3.2 Python (Google Colab): cleaning, filtering, and merging

The WDICSV.csv file, however, presented a different challenge. With approximately 396,970 rows, covering over 1,500 development indicators across more than 200 countries and territories, the dataset was too large for Microsoft Excel to process efficiently. Attempts to filter and sort the data frequently resulted in slow performance, loading failures, or application crashes. To overcome this limitation, the data preparation process was carried out in Python using the pandas library within Google Colab, which is better suited for handling large datasets and enabled the data to be processed efficiently.

The following data preparation steps were completed in Python using the pandas library within Google Colab:

Imported the datasets: Loaded both WDICSV.csv and WDI Country.csv and verified the import process by checking for parsing errors. No errors were identified.

Filtered relevant indicators: Reduced the main dataset from approximately 397,000 rows to a manageable working dataset by filtering for the six development indicators selected for this analysis.

Restricted the analysis period: Limited the dataset to the years 2000–2024. Earlier years contained substantially more missing data, while 2025 data remained incomplete for many countries.

Reshaped the data: Converted the dataset from a wide format (one column per year) to a long format (one row per Country, Indicator, and Year). This structure is better suited for data analysis, visualization, and integration with Power BI.

Handled missing values: Applied a selective approach to missing data rather than using a single rule across all indicators. Indicators with minimal missing values (less than approximately 2%) had incomplete records removed. However, the indicators Population in cities with more than 1 million people (approximately 40% missing) and Population living in slums (% of urban population) (approximately 63% missing) were retained with their missing values intact. Removing these records would have significantly reduced country coverage and introduced bias by disproportionately excluding countries that do not report these statistics.

Integrated country metadata: Merged the cleaned dataset with WDI`Country.csv` using Country Code as the key, adding attributes such as Region and Income Group to support regional and income-based comparisons.

Excluded aggregate records: Separated individual countries from World Bank aggregate entries (e.g., *World*, *Sub-Saharan Africa*, and *Low income*) to ensure that country-level analyses were not distorted by regional or global summary records.

Performed final data validation: Conducted a comprehensive quality check to identify duplicate records, inconsistent country names or country codes, formatting issues (such as leading or trailing whitespace, inconsistent text casing, and numerical formatting inconsistencies), and other potential data integrity issues. No significant issues were identified. What I ended up with was 32,237 country-level rows and 7,053 aggregate-level rows, exported as two CSV files and loaded into Power BI.

4. Analysis Methodology

Once the data was clean, I loaded it into Power BI, where the Value column, set explicitly to Decimal Number, holds the numeric measurement for all six indicators, distinguished by the Indicator Name field.

I defined two calculated columns and one categorical threshold:

- **Has Data**, which flags whether a given row actually has a reported value. I used this to visualize the coverage gaps for the two indicators with high missing values, Slum % and Cities >1M %.
- **Period**, which groups years into 2000 to 2012 and 2013 to 2024, so I could show a simplified before-and-after comparison alongside the year-by-year trend view.
- **Urban Majority**, which classifies each country-year as "Urban Majority" or "Rural Majority" using a 50% urban population threshold. This isn't an official World Bank classification, it's one I defined myself, based on the demographic convention that a country is considered predominantly urban once more than half its population lives in cities.

I also built five core DAX measures to calculate indicator averages filtered by Indicator Name, along with a coverage measure showing the percentage of countries reporting slum data, and used these throughout the dashboards' KPI cards and charts.

From there, I built two dashboards:

- **Dashboard 1, Urbanization Trends**, includes KPI cards for Average Urban Population %, Average Rural Population %, and Average Urban Growth Rate, a multi-line trend chart covering 2000 to 2024, a regional comparison bar chart, a Period-based before-

and-after comparison, and a filled map and donut chart showing the Urban Majority split across countries.

- Dashboard 2, Density and Livability Pressure, includes KPI cards for Average Population Density and Average Cities Over 1M %, a regional density comparison chart, a donut chart showing overall data coverage across all six indicators, a gauge chart showing the share of countries that have reported slum data at least once, and a bar chart of Average Slum % by Income Group.

5. Key Findings

The dashboard analysis revealed several notable patterns and trends in global urbanization:

- **Urban populations now represent the global majority:** On average, 59.43% of the world's population resides in urban areas, compared with 40.57% living in rural areas, indicating that urban living has become the dominant global settlement pattern.
- **Most countries have reached an urban-majority population:** As of 2024, 154 out of 217 countries (approximately 71%) had surpassed the 50% urbanization threshold, meaning that more than half of their population lives in urban areas.
- **Urban population growth has moderated over time:** The average annual urban population growth rate declined from 2.13% during 2000–2012 to 1.73% during 2013–2024. While urbanization continues to increase globally, its rate of growth has slowed in recent years.
- **Significant regional disparities in urbanization remain:** Urban population levels vary considerably across regions. North America recorded the highest average urban population (87.0%), followed by the Middle East and North Africa (72.6%), whereas Sub-Saharan Africa (39.9%) and South Asia (34.3%) remained the least urbanized regions within the dataset.
- **Population density differs substantially between regions:** The global average population density is 411.79 people per square kilometre. However, regional differences are pronounced, with East Asia and the Pacific recording the highest average density (1,031.6 people/km²) and Sub-Saharan Africa the lowest (98.7 people/km²). Despite its relatively low population density, Sub-Saharan Africa also remains the least urbanized region, suggesting that population growth has not yet become as concentrated in urban centres as observed elsewhere.
- **A significant proportion of urban residents live in large metropolitan areas:** On average, 25.06% of a country's population resides in cities with more than one million inhabitants, indicating that a substantial share of urban growth is concentrated in large cities rather than being distributed across smaller urban settlements.

- **Slum population is strongly associated with national income level:** Among all indicators examined, the proportion of the urban population living in slums showed the clearest relationship with income group. On average, slum populations accounted for 65.94% of the urban population in low-income countries, decreasing to 42.62% in lower-middle-income countries, 23.72% in upper-middle-income countries, and only 3.97% in high-income countries. This demonstrates a strong inverse relationship between national income level and the prevalence of informal urban settlements.
- **Data completeness remains an important consideration:** Although approximately 90% of countries reported slum population data for at least one year, only 82% of all possible country-year observations across the six selected indicators contained complete data. This indicates that, while overall coverage is extensive, missing observations remain for specific indicators and years and should be considered when interpreting the results.

6. Insights

What the data reveals:

Overall, the analysis highlights a clear and sustained global transition towards urban living. The majority of the world's population now resides in urban areas, and most countries have surpassed the urban-majority threshold, where more than half of the population lives in cities. However, the pace of urbanization has moderated over time, indicating that although urban growth continues, it is occurring at a slower rate than in previous decades. Furthermore, the distribution of urbanization remains highly uneven across regions. High-income regions, such as North America, have reached very high levels of urbanization, whereas regions including Sub-Saharan Africa and South Asia continue to have predominantly rural populations, reflecting different stages of economic and urban development.

What the patterns suggest:

- The analysis identified a strong and consistent relationship between national income level and the proportion of the urban population living in slums. As countries progress through higher income groups, the prevalence of slum populations declines markedly, falling from approximately 66% of the urban population in low-income countries to less than 4% in high-income countries. This pattern suggests that the quality of urban living conditions is closely associated with a country's level of economic development, highlighting that successful urbanization depends not only on population growth but also on sustained investment in housing, infrastructure, and public services.
- The analysis also highlighted important disparities in data availability, particularly for indicators relating to slum populations and large metropolitan areas. Rather than representing only a data limitation, these gaps may themselves be informative. Countries experiencing rapid and unplanned urbanization may have more limited statistical capacity to collect and report development data consistently, resulting in lower data completeness. Consequently, the regions facing the greatest urban development challenges may also be

those with the least comprehensive evidence base, creating potential blind spots for policymakers and development agencies seeking to address urban livability and infrastructure needs.

7. Recommendations

- **Prioritize proactive urban planning in emerging urban-majority countries:** Countries approaching the 50% urban-majority threshold, particularly in Sub-Saharan Africa and South Asia, should invest in housing, transport infrastructure, and essential public services before rapid urbanization places significant pressure on existing systems. Proactive planning is likely to be more effective and cost-efficient than responding to challenges after they emerge.
- **Target affordable housing initiatives in lower-income economies:** Given the strong inverse relationship observed between national income level and slum population prevalence, governments and development partners should prioritize affordable housing, basic infrastructure, and urban renewal programmes in low-income and lower-middle-income countries, where the proportion of the urban population living in slums remains highest.
- **Strengthen the quality and consistency of urban development data:** International development organizations and national statistical agencies should work together to improve the collection and reporting of indicators relating to slum populations and large metropolitan areas. The presence of missing country-year observations limits the ability of policymakers to accurately monitor urban living conditions and identify emerging areas of concern.
- **Monitor rapidly urbanizing regions closely:** Regions such as Sub-Saharan Africa, which currently exhibit relatively low levels of urbanization but substantial population growth, should be closely monitored. As urbanization accelerates, these regions may experience increasing demand for housing, infrastructure, and public services, making early planning essential to support sustainable urban development.
- **Adopt region-specific and income-sensitive policy approaches.** Policymakers should avoid relying solely on global averages when setting urban development targets. The analysis demonstrates substantial variation across both regions and income groups, highlighting the need for strategies that reflect local development contexts, demographic trends, and urbanization patterns.

8. Conclusion

This project gave me the opportunity to complete a full data analytics workflow, from collecting and cleaning a large real world dataset to building an interactive Power BI dashboard and turning

the results into meaningful insights. By cleaning, analysing, and visualising the data, I was able to answer key questions about global urbanization and present the findings in a clear and interactive way.

The analysis showed that global urbanization is a long-term trend, with most of the world's population now living in urban areas. However, the results also showed that urbanization varies greatly across regions and income groups. One of the strongest findings was the relationship between income level and slum population, with higher-income countries generally having much lower slum populations. The project also highlighted that missing data can be an important finding in itself, as gaps in reporting make it harder to understand urban development in some parts of the world.

Working with a dataset of this size also showed the importance of choosing the right tools for the job. While Excel was useful for exploring the smaller supporting datasets, the main dataset was too large to work with efficiently. Using Python and the pandas library made it much easier to clean and prepare the data before importing it into Power BI for analysis and visualization.

Overall, this project strengthened my skills in data cleaning, data analysis, and dashboard development. More importantly, it showed how data analytics can be used to better understand real-world development challenges and provide insights that can support better planning and decision making.